ASIAN INSTITUTE OF MANAGEMENT

Data Mining and Wrangling

Dimensionality Reduction

Session 17 and 18

BSDSBA 2028

25 March 2026

ASIAN INSTITUTE OF MANAGEMENT

Class Administrative Matters

Course Deliverables

Assessed – R06

Due this Week – ICA07(Wed March 25), E5 (Fri March 27), A1 (Fri March 27),

Upcoming – R08 (Wed March 25), ICA08 (Wed April 1), E6 (Mon April 6),

Mini Project 1 (Mon March 30)

For Checking – ICA04, ICA05, ICA06, E3, Midterm Exam, R07, E4, E3S2

Class Administrative Matters

Course Schedule

April 1 (Sessions 19 and 20) – schedule is moved to April 1 9:00 AM to 12:00 NN

Session 17 and 18 – Information Retrieval

Gameplan

Session 17

Dimensionality Reduction

Principal Component Analysis (PCA)

Session 18

Applied Example of PCA

Introduction to Dimensionality Reduction

Introduction to Dimensionality Reduction

Why do Dimensionality Reduction?

Interpretability – high dimensional data is hard to interpret.

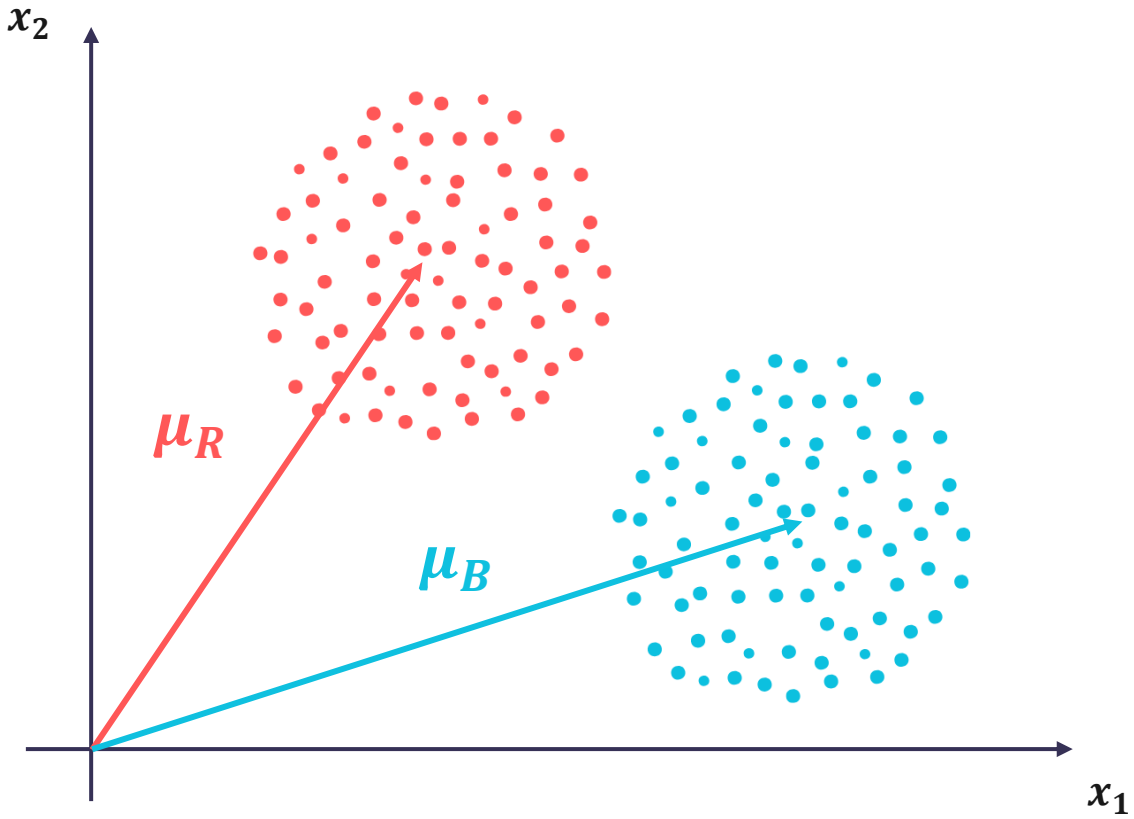
Curse of Dimensionality – as dimensions increase, distances between points become less informative.

Improved Model Performance – reducing dimensions can improve learning.

Computational Efficiency – less features means faster computation and less storage.

Introduction to Dimensionality Reduction

The Curse of Dimensionality



$$\text{Contrast} = \frac{\text{std}_\text{i}(\mu_i)}{\text{average}_i(\mu_i)}$$

Introduction to Dimensionality Reduction

How do we reduce dimensions?

Domain Knowledge – drop features known to be irrelevant

Empirical Feature Selection – train models using subsets of features.

Removing Noise – removing irrelevant or non-informative features.

Reducing Redundancy – removing a correlated feature.

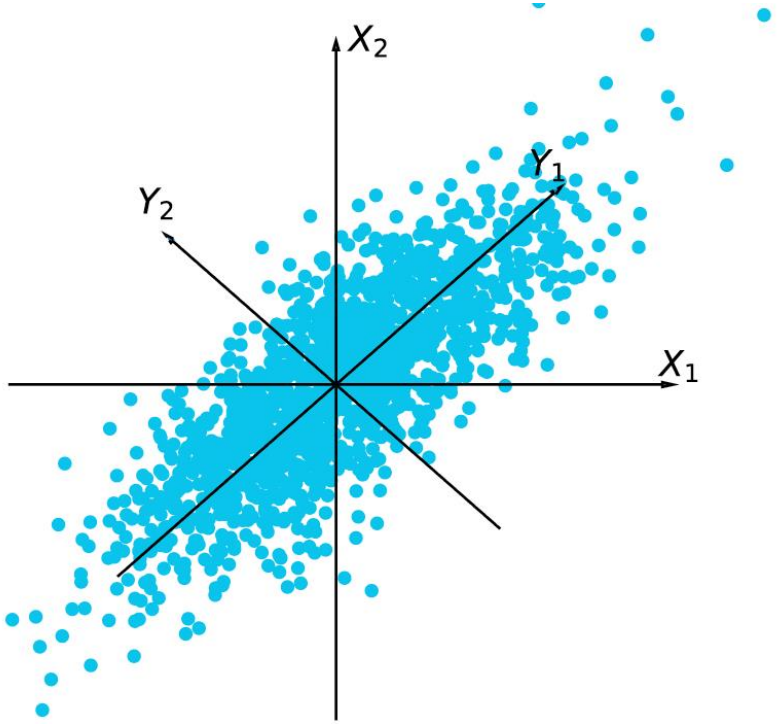
Feature Engineering – creating new features to compress information.

Principal Component Analysis (PCA)

Principal Component Analysis (PCA)

The Core Idea of PCA

The goal of PCA is to find a rotated axis-system where the greatest amount of variance is captured in a small number of dimensions.



Principal Component Analysis (PCA)

PCA Prerequisites

Vectors

Vectors represent a collection of numbers arranged in a specific order.

$$\vec{x} = [x_1, x_2, \dots, x_n]$$

Dot Product

The dot product describe the projection of one vector with another vector.

$$\vec{x} \cdot \vec{y} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

Principal Component Analysis (PCA)

PCA Prerequisites

Matrix

A two-dimensional array (or table) of numbers arranged in rows and columns.

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$$

A matrix can act onto and transform vectors.

$$\vec{\boldsymbol{x}}' = \boldsymbol{A}\vec{\boldsymbol{x}}$$

Principal Component Analysis (PCA)

Example: Vector Transformation 1

Transform the vector $\vec{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ using the matrix $\boldsymbol{A} = \begin{bmatrix} -5 & 2 \\ -7 & 4 \end{bmatrix}$

Principal Component Analysis (PCA)

Example: Vector Transformation 2

Transform the vector $\vec{x} = \begin{bmatrix} 2 \\ 7 \end{bmatrix}$ using the matrix $\boldsymbol{A} = \begin{bmatrix} -5 & 2 \\ -7 & 4 \end{bmatrix}$

Principal Component Analysis (PCA)

PCA Prerequisites

Eigenvalue and Eigenvector

Let A be an $n \times n$ matrix and let $\vec{x}$ be a non-zero vector for which

$$A\vec{x} = \lambda\vec{x}$$

for some scalar λ. Then λ is called an **eigenvalue** of matrix A and $\vec{x}$ is an **eigenvector** of A associated with λ.

Principal Component Analysis (PCA)

PCA Prerequisites

The Existence of Eigenvector

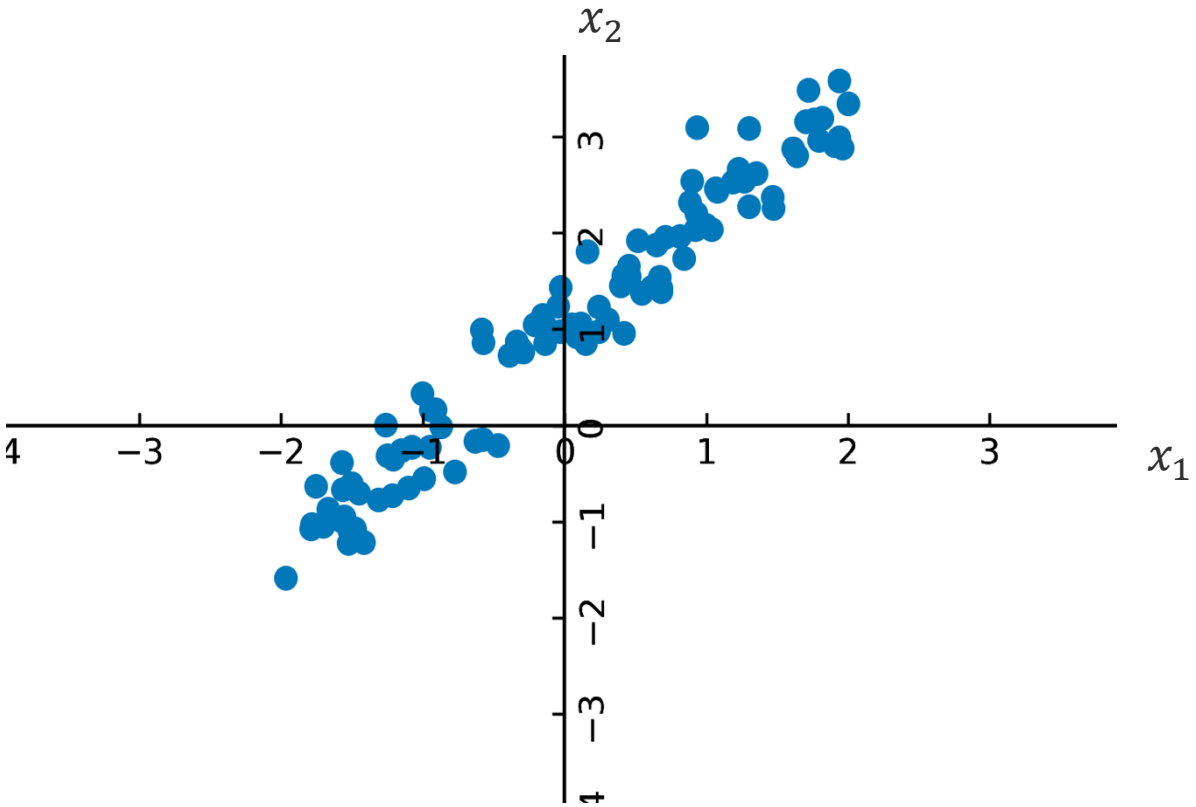
Let A be an $n \times n$ matrix then suppose $\det(\lambda I - A) = 0$ for some λ. Then λ is an eigenvalue of A and there exists a nonzero vector $\vec{x}$ such that

$$A\vec{x} = \lambda\vec{x}$$

Principal Component Analysis (PCA)

Derivation of PCA

Given a p-dimensional feature vectors, we want to summarize them by projecting down into a q-dimensional subspace by finding a projection or transformed axes which maximizes the variance.



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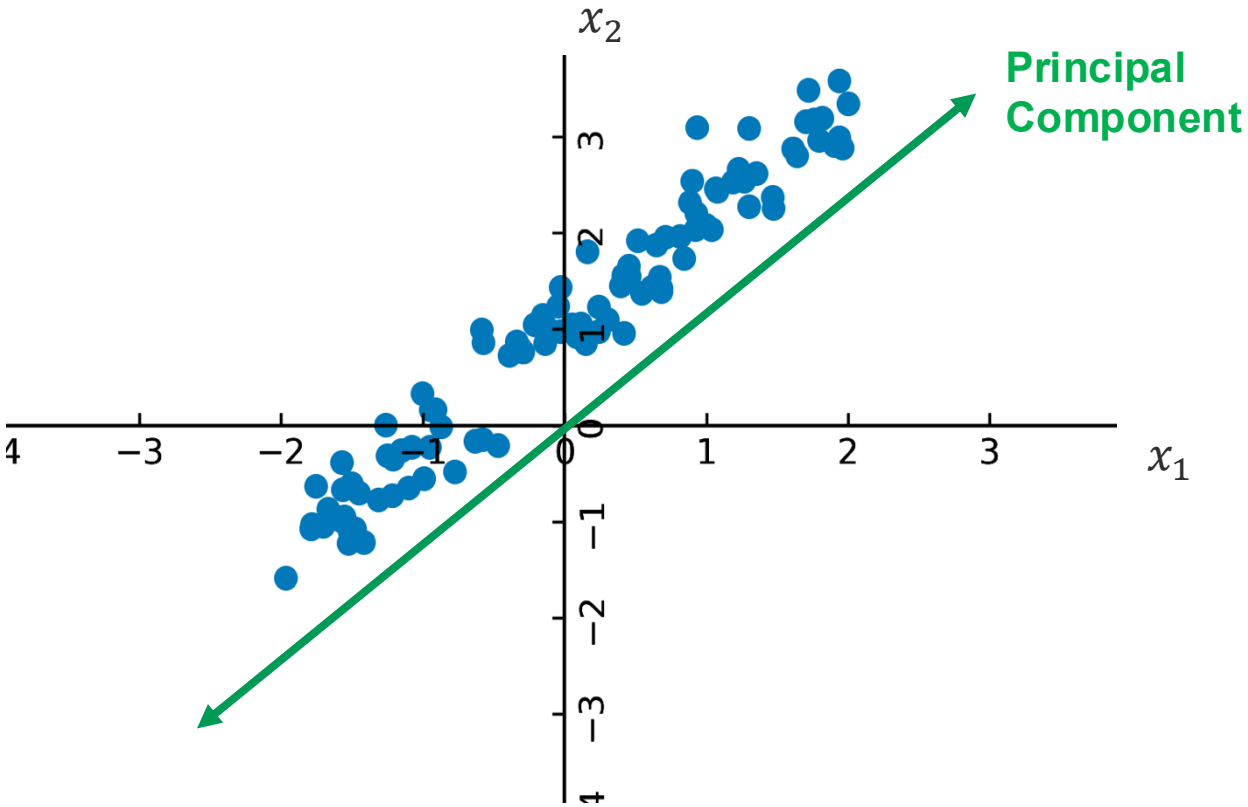

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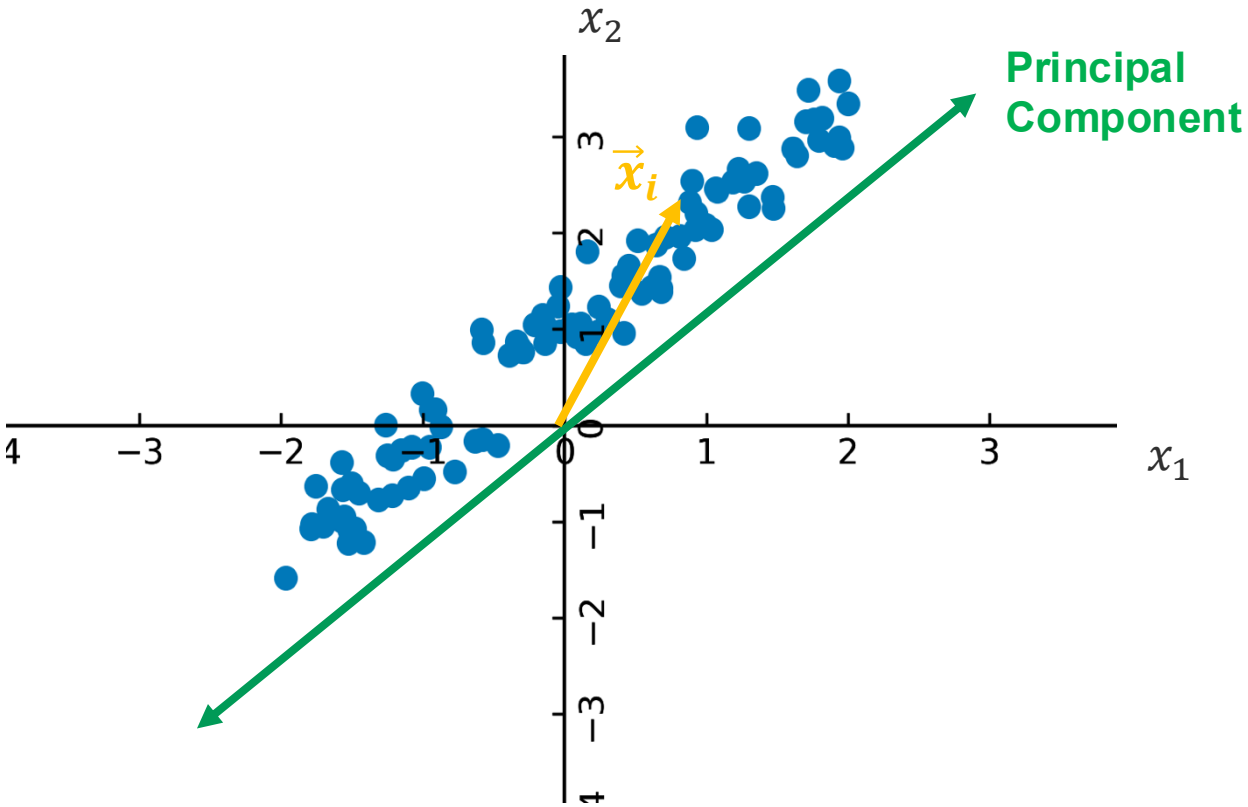

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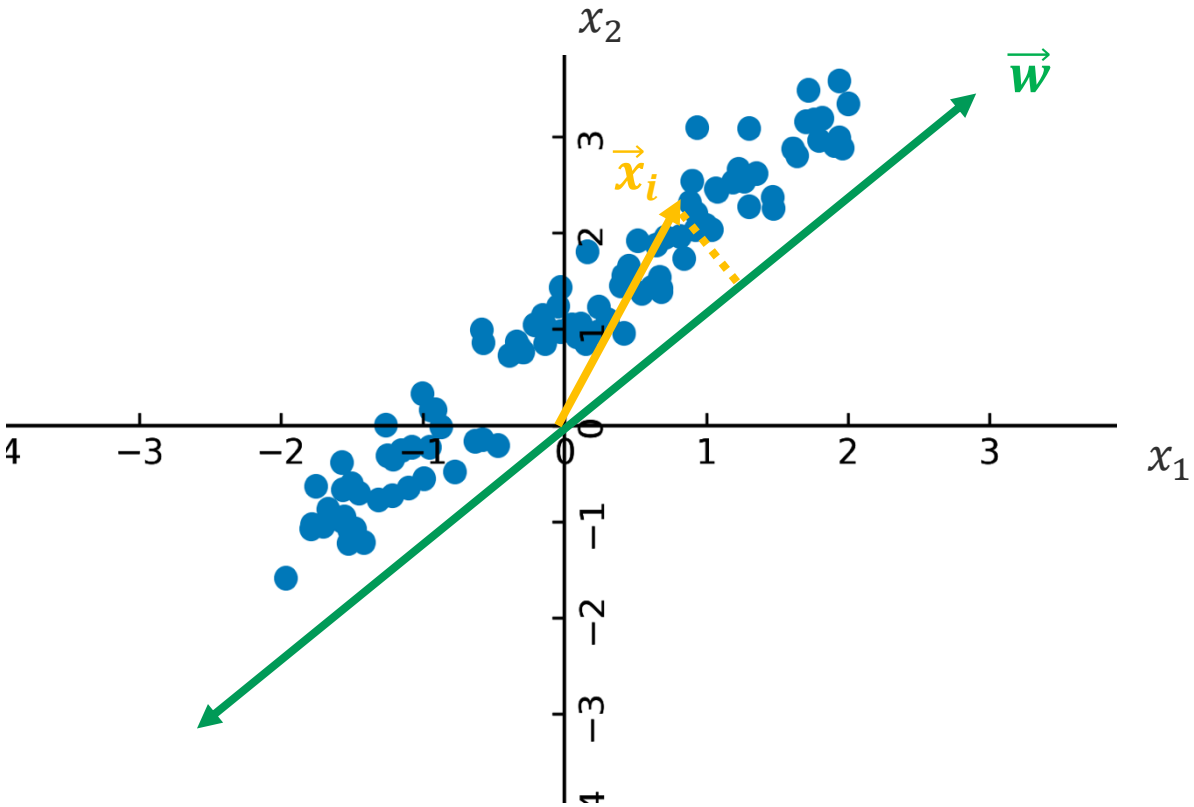

Principal Component Analysis (PCA)

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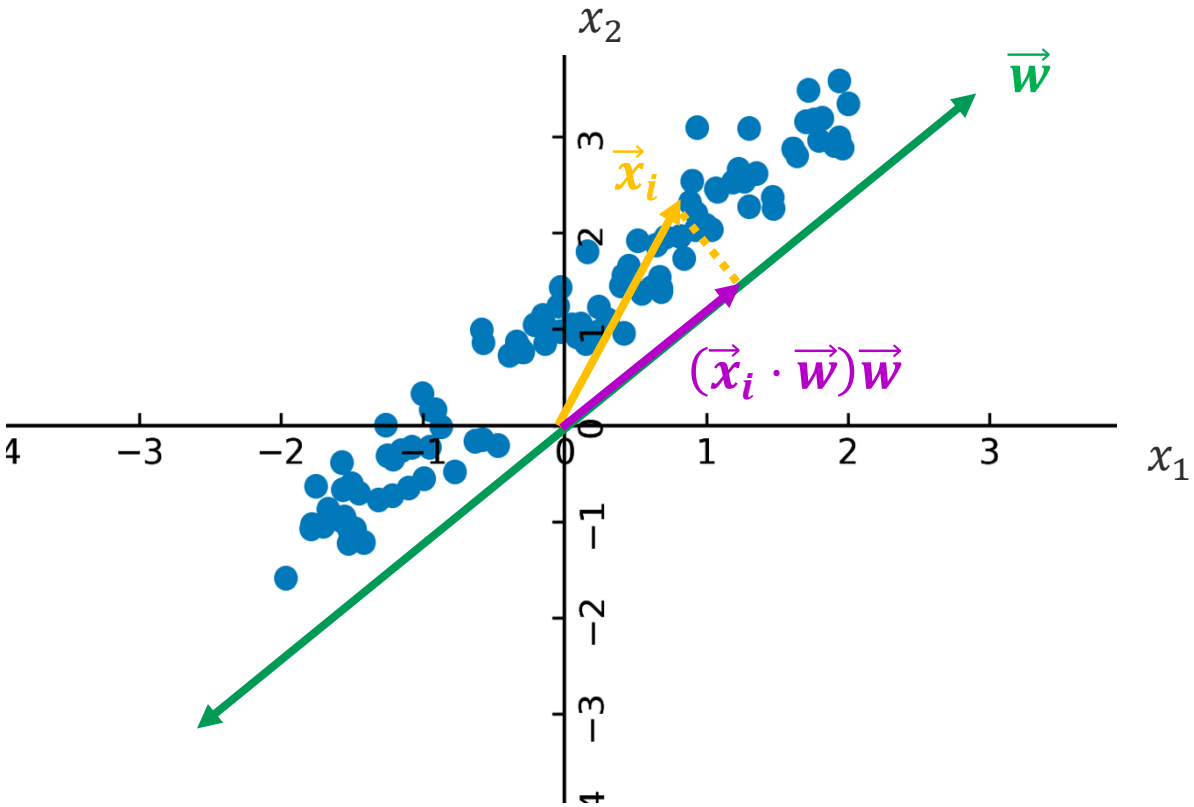
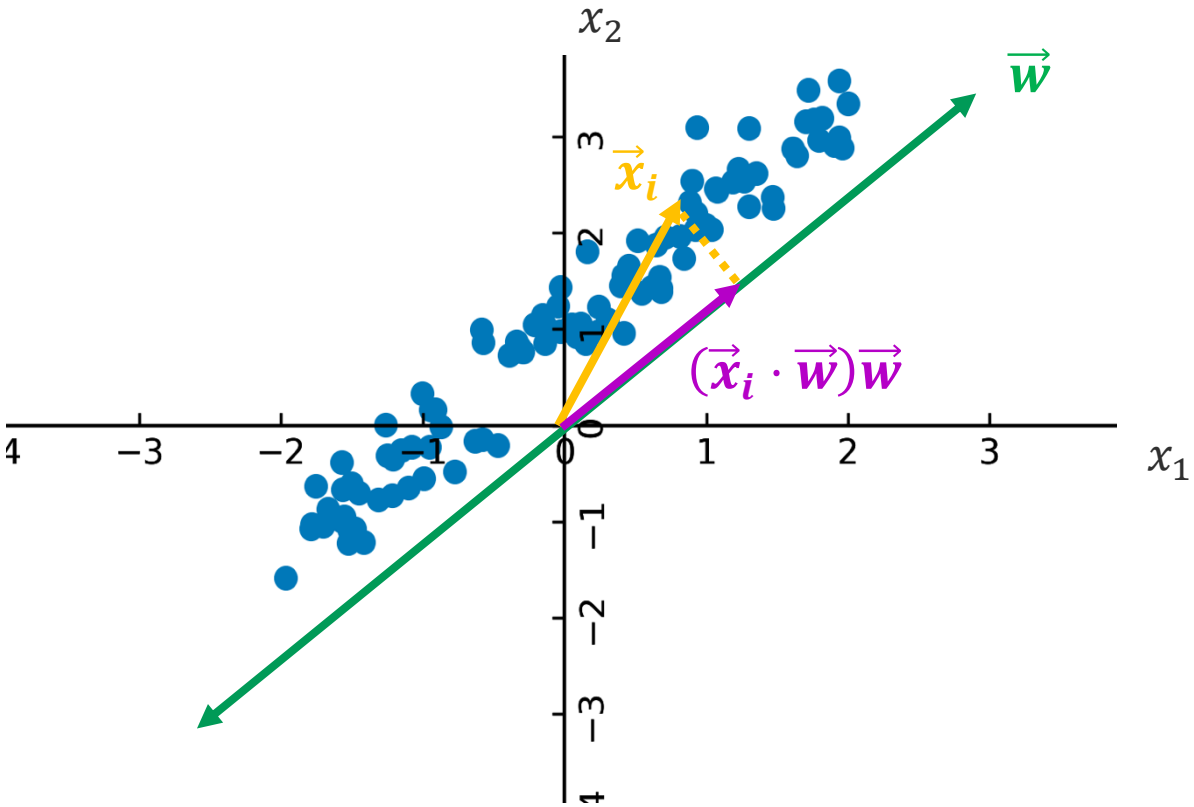


Principal Component Analysis (PCA)

Derivation of PCA

We reframe the problem to: Finding $\vec{w}$ such that projections of the vectors $\vec{x}_i$ onto $\vec{w}$ are maximized.



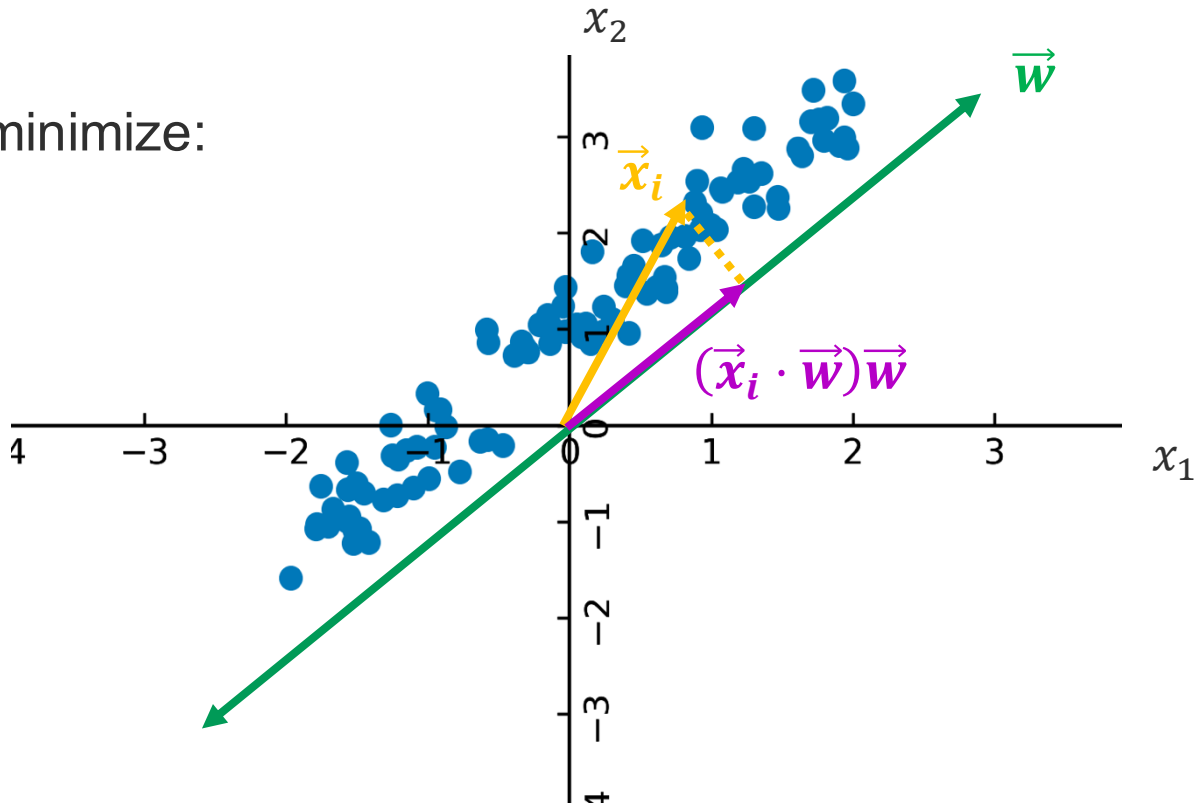
Principal Component Analysis (PCA)

Derivation of PCA

We reframe the problem to: Finding $\vec{w}$ such that projections of the vectors $\vec{x}_i$ onto $\vec{w}$ are maximized.

Hence, we want to minimize:

$$\|\vec{x}_i - (\vec{x}_i \cdot \vec{w})\vec{w}\|^2$$



Principal Component Analysis (PCA)

$$\|\vec{x}_i - (\vec{x}_i \cdot \vec{w})\vec{w}\|^2$$

Principal Component Analysis (PCA)

Find $\vec{w}$ such that the projection residual $\sum_i^n [\|\vec{x}_i\|^2 - (\vec{x}_i \cdot \vec{w})^2]$ is minimized

Principal Component Analysis (PCA)

$$\frac{1}{n}\sum_{i=1}^{n}(\vec{x}_i \cdot \vec{w})^2 =$$

Principal Component Analysis (PCA)

Maximize $\vec{w}^T \boldsymbol{V} \vec{w}$

Principal Component Analysis (PCA)

$$\frac{\partial}{\partial \vec{w}}[\vec{w}^T V \vec{w}]$$

Principal Component Analysis (PCA)

Maximize $\vec{w}^T \boldsymbol{V} \vec{w}$

$$\frac{\partial u}{\partial \vec{w}} = 2\boldsymbol{V}\vec{w} - 2\lambda\vec{w} = 0$$

$$\boldsymbol{V}\vec{w} = \lambda\vec{w}$$

The **Principal Components** $\vec{w}$ are ***eigenvectors*** of the **covariance matrix** $\boldsymbol{V}$ and the maximizing vector will be the one associated with the largest eigenvalue λ**.**

Principal Component Analysis (PCA)

Variance Explained

Given a p-dimensional feature vectors, we want to summarize them by projecting down into a q-dimensional subspace by finding a projection or transformed axes which maximizes the variance.

Variance explained of the *i*-th PC $= \frac{\lambda_i}{\sum_j^p \lambda_j}$

Cumulative variance explained of selected *q* PCs $= \frac{\sum_j^q \lambda_i}{\sum_j^p \lambda_j}$

Principal Component Analysis (PCA)

Interpreting the PCs

Coordinates – Draw arrows that show the original feature vector coordinates direction with respect to each PCs obtained.

Correlations – Arrows clusters into groups of highly related attributes.

Clusters – Describe the clusters of point observed with respect to the PCs

Funnels – Funnels are wide at one end and narrow at the other. They happen when one PC affects the variance of another PC.

Voids – Areas which are unpopulated.

Principal Component Analysis (PCA)

A Recipe for Applying PCA

1. **Normalize the input data** – ensure attributes with large values will not dominate attributes with smaller values.
2. **Compute the PCs** – find directions that capture most variance in data.
3. **Decide how many PCs to retain** – choose how many to keep while preserving most of the information.
4. **Transform the data onto the PCs** – transform the original data into the reduced-dimensional space.
5. **Visualize and Interpret the PCs** – understand what the components reveal about the data.

Principal Component Analysis (PCA)

Limitations of PCA

- **PCA** cannot capture nonlinear relationships.
- A core assumption of **PCA** is that high variance means important information.
- **PCs** can reduce interpretability
- **PCA** is sensitive to scaling
- **PCA** is an unsupervised global method